

Solving Dimensionality Problems in Heterogenous Agent Models with Perturbation Methods (Bilal 2023, Bilal and Rossi-Hansberg 2023)

Berkay Saygin¹

¹University of Chicago

March 10, 2026

Outline

- **MOTIVATION:** Solving dynamic models with large state spaces
 - a frictional migration model with > 300 locations
 - agents that are heterogeneous in wealth that takes continuous values
- **APPROACH:** Combine HJB + KFE into a Master equation, linearize around steady state
- **GOAL:** Represent coupled equations as one and Bellmanize it (state space representation)
- This session builds on:
 - 1 Bilal (2023), “Solving Heterogeneous Agent Models with the Master Equation”
 - 2 Bilal and Rossi-Hansberg (2023), “Anticipating Climate Change Across the United States”

Overview

- 1 Motivation
- 2 Mathematical Preliminaries
- 3 Deriving the Master Equation (Bilal 2023)
- 4 Application: Bilal & Rossi-Hansberg (2023)

Motivation from Bilal and Rossi-Hansberg (2023)

- Core question: How does climate change affect agents' location decisions?
- This requires a model that:
 - Permits heterogeneous climate impacts across space (e.g. inland vs. coastal Florida)
 - Captures forward-looking behavior (agents anticipate future climate dynamics)
 - Allows adaptation through both migration and capital accumulation

Choosing a Model and a Solution Method

- **Desmet, Nagy, Rossi-Hansberg (2018):**
 - Pro: Straightforward to solve
 - Con: No forward-looking behavior, no capital accumulation
- **Caliendo, Dvorkin, Parro (2019); Kleinman, Liu, Redding (2023):**
 - Pro: Forward-looking behavior and capital accumulation
 - Con: Solution methods limited to relatively few locations (~ 50)
- **Conclusion:** A different approach is needed — one that handles large state spaces while preserving forward-looking dynamics

Solution: Continuous Time + Perturbation Methods

- Formulate the model in continuous time with many locations (e.g., $\approx 3,000$ counties)
- Solve via perturbation around a steady state:
 - Analogous to linearizing a representative-agent model (e.g., RBC)
 - Extended to accommodate heterogeneous agents and a high-dimensional state space
- Focus: the first-order (linear) response to aggregate shocks

Solution: Steps

OVERALL STRATEGY (STARTING FROM A STEADY STATE):

- 1 Express prices and wages as functions of the distribution of agents across locations
- 2 Combine the Hamilton-Jacobi-Bellman (HJB) equation and the Kolmogorov Forward Equation (KFE) into a single “Master Equation”
- 3 Linearize (perturb) the Master Equation around the steady state
- 4 Solve the linearized system via undetermined coefficients to recover the value function's response
- 5 Use the linearized KFE to track how the distribution of agents evolves under shocks

KEY OUTCOME: A tractable characterization of transitional dynamics across a very large number of locations.

Intuition: Why the Master Equation?

- **THE COMPUTATIONAL CHALLENGE:** With 3,000 counties, the HJB (backward) depends on the future path of N , while the KFE (forward) depends on the current value function V . Solving this coupled system by iterating on entire time paths scales poorly.
- **KEY INSIGHT:** The Master Equation encodes both individual optimization (HJB) and aggregate dynamics (KFE) in a single equation by treating the distribution N as an explicit state variable.
- **PERTURBATION:**
 - Analogous to linearizing an RBC model around steady state to obtain impulse responses
 - Here, the same logic applies but with a distribution as the state variable
 - A small shock $\Rightarrow$ how does the distribution of agents across locations shift?
- **RESULT:** The FAME (First-order Approximation to the Master Equation) reduces the coupled forward-backward system to finite-dimensional matrix equations, eliminating the need for iterative computation.

Mathematical Preliminaries

Functions, Operators, and Functionals

- **FUNCTION:** Maps numbers to numbers.
- **OPERATOR:** Maps functions to functions. E.g. $\mathcal{A}[f](x) = f''(x)$, or the HJB operator $V \mapsto \rho V - u - \mathcal{A}[V]$
- **FUNCTIONAL:** Maps a function to a number. E.g. $\mathcal{F}[f] = \int_0^1 f(x) dx$, or aggregate welfare $\mathcal{W}[V] = \int V_i dN_i$
- **Key observation:** In Bilal (2023), the value function V depends on the distribution N . In models with a continuous state space, V is a functional of $N(\cdot)$ and differentiating it requires Fréchet calculus. In the discrete spatial model, N is a finite vector and standard partial derivatives suffice.

From Gradients to Functional Derivatives

FINITE DIMENSIONS: For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the directional derivative in direction h :

$$f(z + \epsilon h) - f(z) \approx \epsilon \sum_{i=1}^n \frac{\partial f}{\partial z_i} h_i = \epsilon \langle \nabla f, h \rangle$$

- The gradient ∇f collects all partial derivatives; its inner product with h yields the rate of change along h

INFINITE DIMENSIONS: For a functional $V[g]$, the analogous expression:

$$V[g + \epsilon \delta g] - V[g] \approx \epsilon \int \frac{\partial V}{\partial g}(x, x', g) \delta g(x') dx'$$

- $\frac{\partial V}{\partial g}(x, x', g)$: the functional gradient, encoding sensitivity of V at x to perturbations of g at x'
- $\delta g(x')$ is the perturbation direction (an arbitrary function); $\int dx'$ replaces $\sum_i$

Recovering a Partial Derivative

Setting $\delta g = \delta_{x'_0}$ (a Dirac delta at a single point x'_0):

$$\int \frac{\partial V}{\partial g}(x, x', g) \delta_{x'_0}(x') dx' = \frac{\partial V}{\partial g}(x, x'_0, g)$$

⇒ This isolates sensitivity at a single point — the analog of a single partial derivative $\frac{\partial f}{\partial z_i}$ in finite dimensions

- In the discrete spatial model: setting the perturbation to a single location j recovers $\frac{\partial V_i}{\partial N_j} = v_{ij}$

The Fréchet Derivative

The previous slides showed the directional application of a functional derivative — how V changes along a specific perturbation δg . The **Fréchet derivative** is the stronger underlying object:

- $D\mathcal{F}|_g$ is a bounded linear map such that:

$$\mathcal{F}[g + h] = \mathcal{F}[g] + D\mathcal{F}|_g[h] + o(\|h\|) \quad \text{for all } h$$

- The approximation must hold uniformly across all directions h , with the remainder vanishing faster than $\|h\|$
- By the Riesz representation theorem (in Hilbert spaces), $D\mathcal{F}|_g[h] = \langle \frac{\delta \mathcal{F}}{\delta g}, h \rangle$, so the Fréchet derivative can be identified with the functional gradient $\frac{\delta \mathcal{F}}{\delta g}$

Interpretation: The Fréchet derivative is the “gradient in function space” — it characterizes the best linear approximation to $\mathcal{F}$ at g , just as ∇f does in $\mathbb{R}^n$.

Fréchet Derivatives: Discrete Intuition

- In the spatial model, $N = (N_1, \dots, N_I)$ is a finite-dimensional vector
- The Fréchet derivative therefore reduces to the **Jacobian**:

$$V_i(N + \epsilon n) \approx V_i(N) + \epsilon \sum_{j=1}^I \underbrace{\frac{\partial V_i}{\partial N_j}}_{\equiv v_{ij}} n_j$$

- v_{ij} measures the marginal effect on the value of residing in location i when the population in location j increases
- This is precisely the matrix $v \in \mathbb{R}^{I \times I}$ that the FAME solves for
- **Summary:** In discrete space, the Fréchet derivative reduces to standard partial derivatives collected in a matrix. No functional analysis is required.

The Coupled System: HJB and KFE in Operator Notation

The two equations that govern the model in continuous time:

HJB (backward — individual optimization):

$$\rho V_t(x) = u(x, g_t) + \mathcal{M}[V_t](x) + \frac{\partial V_t}{\partial t}(x)$$

- u : flow payoff (depends on the distribution g_t); $\mathcal{M}$: generator (migration/diffusion option value); $\frac{\partial V}{\partial t}$: continuation value

KFE (forward — distributional dynamics):

$$\frac{\partial g_t}{\partial t}(x) = \mathcal{M}^*[g_t](x)$$

- $\mathcal{M}^*$: the adjoint (Fokker–Planck) operator; governs how the density of agents evolves

THE PROBLEM: The HJB depends on the future path of g (through $\frac{\partial V}{\partial t}$); the KFE depends on the current V (through $\mathcal{M}^*$). They are coupled.

Anticipating the Construction of the Master Eq. with Fréchet Chain Rule

GOAL OF THE MASTER EQ.: Eliminate $\frac{\partial V}{\partial t}$ by expressing it in terms of V and g alone.

- If V depends on a time-varying distribution $g_t(\cdot)$, the chain rule yields:

$$\frac{\partial V_t}{\partial t}(x) = \int \frac{\partial V}{\partial g}(x, x', g_t) \frac{\partial g_t}{\partial t}(x') dx' \equiv \left\langle \frac{\partial V}{\partial g}(x, g_t), \frac{\partial g_t}{\partial t} \right\rangle$$

- Same structure as the finite-dimensional chain rule $\frac{dV_i}{dt} = \sum_j \frac{\partial V_i}{\partial N_j} \frac{\partial N_j}{\partial t}$, with $\int dx'$ replacing $\sum_j$

CONSTRUCTION OF THE MASTER EQUATION:

- 1 The HJB contains $\frac{\partial V}{\partial t}$
- 2 Apply the chain rule above to express $\frac{\partial V}{\partial t}$ in terms of $\frac{\partial V}{\partial g}$ and $\frac{\partial g}{\partial t}$
- 3 Substitute the KFE for $\frac{\partial g}{\partial t}$, so that everything is expressed in terms of V and g
- 4 The result is a single, fully recursive equation — the **Master Equation**

When Are Fréchet Derivatives Necessary?

CONTINUOUS STATE SPACE (e.g. wealth in Aiyagari/Bewley/Hugett):

- The distribution $g(a)$ is a density on $[\underline{a}, \infty)$, $V[a, g]$ is a functional
- The functional derivative $\left. \frac{\delta V}{\delta g} \right|_g(a, a')$ is a kernel in two continuous variables: the agent's own state a and the perturbation point a'
- Fréchet calculus is required; resulting equations are discretized for numerical computation

DISCRETE STATE SPACE (e.g. I locations in the spatial model):

- Distribution $N = (N_1, \dots, N_I) \in \mathbb{R}^I$ is a vector, $V(z, N)$ is a function $\mathbb{R} \times \mathbb{R}^I \rightarrow \mathbb{R}$
- Differentiating with respect to N yields ordinary partial derivatives $\frac{\partial V_i}{\partial N_j} = v_{ij}$
- **No functional analysis is required** — standard multivariable calculus suffices

Why Does Bilal Use the Fréchet Framework?

- In the discrete spatial model, no discretization step is needed either (the state space is already discrete)
- The paper covers both continuous and discrete cases
- The Fréchet formulation provides a **unified treatment**; the discrete spatial model is a special case in which the machinery reduces to matrix algebra

What Does FAME Provide in the Discrete Spatial Case?

If the distribution is already a finite vector, why not solve the system directly?

THE DIFFICULTY IS THE FORWARD-BACKWARD COUPLING, NOT THE DIMENSIONALITY:

- HJB (backward): V depends on the future path of N
- KFE (forward): N depends on the current V
- Standard approach: iterate on entire time paths $(V_t, N_t)_{t \geq 0}$ until convergence
- With $I = 3,000$ locations, this requires iterating on 3,000-dimensional trajectories in both temporal directions — computationally prohibitive

FAME ELIMINATES THE ITERATION:

- 1 Solve a Sylvester equation for $v \in \mathbb{R}^{I \times I}$ (a single matrix equation)
- 2 Solve a linear system for $\omega \in \mathbb{R}^I$ (direct computation)
- 3 Integrate the linearized KFE forward (a linear ODE)

COMPARISON: CDP (2019) and KLR (2023) solve the forward-backward system directly, limiting them to ~ 50 locations. FAME handles 3,143 counties on standard hardware.

Adjoint Operators: Definition

- Given a linear operator $\mathcal{A}$, its **adjoint** $\mathcal{A}^*$ is defined by:

$$\langle \mathcal{A}[f], g \rangle = \langle f, \mathcal{A}^*[g] \rangle \quad \text{for all } f, g$$

where $\langle f, g \rangle = \int f(x) g(x) dx$ (or $\sum_i f_i g_i$ in discrete space)

- MATRIX ANALOGY:** If $\mathcal{A}$ is a matrix A , then $\mathcal{A}^*$ is simply A^T (the transpose):

$$\langle Af, g \rangle = \langle f, A^T g \rangle \quad \text{where } \langle f, g \rangle = f^T g$$

- Interpretation:** The adjoint transfers the operator from one argument of the inner product to the other. It is the function-space generalization of matrix transposition.

Generator and Fokker–Planck Operator — No Boundary BM Example

SETUP: An agent's state evolves as a diffusion $dy_t = \mu(y) dt + \sigma(y) dW_t$ (e.g. wealth in a consumption-savings problem with stochastic income) with no boundaries.

Infinitesimal generator $\mathcal{M}$: how fast does the expected value of any smooth function $f(y)$ change?

$$\mathcal{M}[f](y) \equiv \lim_{dt \rightarrow 0} \frac{\mathbb{E}[f(y_{t+dt}) \mid y_t = y] - f(y)}{dt} = \mu(y) f'(y) + \frac{1}{2} \sigma^2(y) f''(y)$$

- Applying Itô's lemma to $f(y_t)$ and taking expectations gives exactly $\mathcal{M}[f]$
- When $f = V$ (the value function), $\mathcal{M}[V]$ captures the option value from future state changes

Its adjoint is the **Fokker–Planck operator $\mathcal{M}^*$** , which answers the dual question: how does the density of agents evolve?

$$\mathcal{M}^*[g](y) = -(\mu(y) g(y))' + \frac{1}{2} (\sigma^2(y) g(y))''$$

- $\mathcal{M}$ governs individual trajectories (backward); $\mathcal{M}^*$ governs the population density (forward)
- The Kolmogorov Forward Equation is $\frac{\partial g}{\partial t} = \mathcal{M}^*[g]$; in steady state $\mathcal{M}^*[g] = 0$

Proof: The Adjoint Identity of the No Boundary BM Example (1/2)

Claim: $\langle \mathcal{M}[V], g \rangle = \langle V, \mathcal{M}^*[g] \rangle$ under the L^2 inner product $\langle f, g \rangle = \int f g dy$.

Expand the left-hand side and treat the two terms separately:

$$\langle \mathcal{M}[V], g \rangle = \int [\mu V' + \tfrac{1}{2}\sigma^2 V''] g dy$$

FIRST TERM: Integrate by parts once to transfer the derivative from V to g :

$$\int \mu V' g dy = \underbrace{[\mu g V]_{-\infty}^{\infty}}_{=0} - \int (\mu g)' V dy$$

The boundary term vanishes because g is a density satisfying $g \rightarrow 0$ at $\pm\infty$.

SECOND TERM: Integrate by parts twice:

$$\int \tfrac{1}{2}\sigma^2 V'' g dy = \underbrace{[\tfrac{1}{2}\sigma^2 g V']_{-\infty}^{\infty}}_{=0} - \int (\tfrac{1}{2}\sigma^2 g)' V' dy$$

Proof: The Adjoint Identity of the No Boundary BM Example (2/2)

Continuing the second integration by parts:

$$-\int (\tfrac{1}{2}\sigma^2 g)' V' dy = \underbrace{\left[-(\tfrac{1}{2}\sigma^2 g)' V\right]_{-\infty}^{\infty}}_{=0} + \int (\tfrac{1}{2}\sigma^2 g)'' V dy$$

COMBINING BOTH TERMS:

$$\langle \mathcal{M}[V], g \rangle = \int \underbrace{\left[-(\mu g)' + \tfrac{1}{2}(\sigma^2 g)''\right]}_{= \mathcal{M}^*[g]} V dy = \langle V, \mathcal{M}^*[g] \rangle \quad \square$$

REMARKS:

- $\mathcal{M}^*$ is defined by the expression that emerges when integration by parts transfers all derivatives from V onto g
- The adjoint is the unique operator satisfying the inner product identity
- **Boundary conditions:** with reflecting barriers (e.g. borrowing constraints), appropriate conditions on V and g are required. Take Fernando's class!

Adjoint Operators: Discrete Spatial Analogy

In the discrete location model, the relevant operator is the **migration matrix** M . If π denotes the matrix of transition probabilities (π_{ij} = prob. of moving $i \rightarrow j$):

$$M = \mu(\pi - I)$$

- **HJB side** (acts on values): $(M V)_i$ computes the expected migration option value from i
- **KFE side** (acts on the distribution): $\dot{N} = M^T N$ governs population flows
- The adjoint is the transpose M^T

SUMMARY:

	HJB (backward)	KFE (forward)
Continuous	$\mathcal{M}[V]$	$\mathcal{M}^*[g]$
Discrete	$M V$	$M^T N$

The Master Equation combines both into a single recursive equation.

Deriving the Master Equation (Bilal 2023)

A Simple Spatial Model (Bilal 2023, §1.6)

- Consider a workers-only economy with I locations and population $N = (N_1, \dots, N_I)$
- Individual problem (HJB):** Worker in location i has value V_i :

$$\rho V_{it} = \underbrace{U_i(N_t)}_{\text{flow utility}} + \underbrace{\mathcal{M}_i[V_t]}_{\text{migration option}} + \underbrace{\dot{V}_{it}}_{\text{continuation}}$$

Flow utility depends on N because N determines local wages, rents, and congestion

- Aggregate dynamics (KFE):** The distribution evolves as workers migrate:

$$\dot{N}_{it} = (M^*(V_t) N_t)_i$$

- THE PROBLEM:** The HJB looks backward (depends on future N); the KFE looks forward (depends on current V). They are coupled.

Step 1: Change of Variables (Bellmanization)

KEY IDEA: Instead of indexing V by calendar time, index it by the state — the distribution N :

$$V_{it} \longrightarrow V_i(N_t)$$

- **Why?** In a standard Bellman equation, the value depends on the current state, not on calendar time. Here, N is the payoff-relevant state (determines all local prices).
- **Consequence:** The continuation value $\dot{V}$ must be re-expressed via the chain rule:

$$\dot{V}_i = \sum_{j=1}^I \frac{\partial V_i}{\partial N_j} \dot{N}_j$$

In words: the value of location i changes over time because the distribution N changes over time. The rate of change equals the sensitivity of V_i to each N_j , times the rate at which N_j moves.

Step 2: Plug the KFE into the Chain Rule

We know from the KFE that $\dot{N}_j = (M^*(V)N)_j$. Substitute into the chain rule:

$$\dot{V}_i = \sum_{j=1}^I \frac{\partial V_i}{\partial N_j} \underbrace{\dot{N}_j}_{=(M^*(V)N)_j} = \sum_{j=1}^I \frac{\partial V_i}{\partial N_j} (M^*(V)N)_j$$

- **What happened?** The KFE — which governed the evolution of N as a separate equation — has been absorbed into the expression for $\dot{V}$.
- The continuation value now depends only on V (through migration shares in M^*) and N (the current population vector)
- No separate forward equation for N remains

Step 3: The Master Equation

Substitute $\dot{V}_i$ back into the HJB:

$$\rho V_i(N) = U_i(N) + \mathcal{M}_i[V(N)] + \sum_j \frac{\partial V_i}{\partial N_j} (M^*(V(N)) N)_j$$

- This is the **Master Equation**: a single, self-contained, recursive equation
- **WHAT WAS GAINED:**
 - Before: two coupled equations (HJB backward, KFE forward) requiring iterative solution on entire time paths
 - After: one Bellman equation in (V, N) at a single point in time
- Three terms: (1) $U_i(N)$: flow payoff; (2) $\mathcal{M}_i[V]$: migration option value; (3) $\sum_j \frac{\partial V_i}{\partial N_j} (M^* N)_j$: continuation value from how migration reshapes N

Step 4: Linearize Around Steady State

The Master Equation is nonlinear and high-dimensional. Solve it by perturbation:

- **Steady state:** Set all time derivatives to zero:

$$\rho V_i^{SS} = U_i(N^{SS}) + \mathcal{M}_i[V^{SS}], \quad M^*(V^{SS})N^{SS} = 0$$

The continuation term vanishes because the distribution is stationary

- **Perturb:** Set $N = N^{SS} + \epsilon n$ and seek a first-order expansion:

$$V_i(N^{SS} + \epsilon n) = V_i^{SS} + \epsilon \sum_j v_{ij} n_j + O(\epsilon^2)$$

where $v_{ij} \equiv \left. \frac{\partial V_i}{\partial N_j} \right|_{SS}$ is the **impulse value**: how the value of location i responds to a marginal increase in N_j

- **Substitute** the expansion into the Master Equation and collect first-order terms in ϵ

Step 4b: Differentiating the Master Equation

Differentiate the Master Equation (slide 29) w.r.t. N_j , evaluating at SS. Each term on the RHS produces a term in the FAME:

TERM 1: $U_i(N) \rightarrow \frac{\partial U_i}{\partial N_j} \Big|_{SS} \equiv D_{ij}$ (direct price impact)

TERM 2: $\mathcal{M}_i[V(N)] \rightarrow \sum_k \frac{\partial \mathcal{M}_i}{\partial V_k} \Big|_{SS} \cdot v_{kj} = (Mv)_{ij}$

- By the envelope theorem, differentiating through the log-sum-exp uses the steady-state migration shares — only M at SS matters

TERM 3: Differentiate $\sum_k \frac{\partial V_i}{\partial N_k} (M^*(V)N)_k$ w.r.t. N_j by the product rule:

- $\frac{\partial V_i}{\partial N_k}$ (yielding second derivatives $\frac{\partial^2 V_i}{\partial N_j \partial N_k}$) is multiplied by $(M^*N)_k \Big|_{SS} = 0$, so it **drops out**
- What remains: v_{ik} times the derivative of the LOM, which depends on N_j through **two** channels:
 - Direct (through N): $v \cdot \frac{\partial (M^*N)}{\partial N_j} \Big|_{SS} = (vM^*)_{ij}$ (population propagation)
 - Indirect (through $V(N)$): $v \cdot \frac{\partial (M^*(V)N)}{\partial V} \Big|_{SS} \cdot v_{.j} = (vGv)_{ij}$ (behavioral feedback)

Step 5: The FAME (First-Order Approximation)

Collecting first-order terms yields a **matrix equation** for $v \in \mathbb{R}^{I \times I}$:

$$\rho v = D + Mv + vM^* + vGv$$

This is a (nonlinear) **Sylvester equation**. Each term has economic content:

- ➊ D : **direct price impact** — when N_j increases, local wages and rents adjust, shifting V_j
- ➋ Mv : **partial equilibrium migration** — workers re-evaluate the migration option using the steady-state matrix M . By the envelope condition, only M at SS matters.
- ➌ vM^* : **GE force 1** — an additional worker in j migrates like any other worker (M^* is the KFE operator), redistributing population and thereby affecting V_j
- ➍ vGv : **GE force 2** — the additional worker changes prices, which changes everyone's migration decisions, feeding back through the law of motion

Step 6: Transitional Dynamics

Given the FAME solution v , transitional dynamics follow immediately:

$$\dot{n}_t = (M^* + Gv) n_t$$

- This is a **linear ODE**: given any initial perturbation n_0 , the population response is computed by matrix exponentiation — no iteration required

SUMMARY OF THE PROCEDURE:

- 1 Solve the nonlinear steady state (V^{SS}, N^{SS})
- 2 Solve the Sylvester equation for v (the FAME)
- 3 Integrate the linear ODE forward for transitional dynamics

Looking Ahead

- The same logic extends to richer models
- Adding capital, capitalists, and climate shocks introduces more state variables (N, K) and a second HJB (for capitalists)
- But the five-step recipe remains identical:
 - 1 Write HJB & KFE
 - 2 Change variables
 - 3 Plug in laws of motion
 - 4 Obtain the Master Equation
 - 5 Linearize

Repeating the Recipe with a Richer Model

- We now apply the same five-step procedure from Bilal (2023) to the full model of Bilal & Rossi-Hansberg (2023)
- WHAT CHANGES RELATIVE TO THE SIMPLE CASE:
 - Two agent types: workers (mobile) and capitalists (immobile)
 - Two state variables: population N_t and capital K_t (total: $> 6,000$ dimensions)
 - Two HJB equations: worker (eq. 3) and capitalist (eq. 7)
 - Two laws of motion: KFE for N (eq. 5) and capital accumulation for K (eq. 8)
 - Climate shocks affecting productivity, amenities, and depreciation

What Stays the Same

- THE PROCEDURE IS IDENTICAL:

- 1 Change of variables
- 2 Apply chain rule
- 3 Plug in laws of motion
- 4 Obtain the Master Equation
- 5 Linearize $\rightarrow$ FAME

Model: Overview (Bilal & Rossi-Hansberg 2023, §2)

- $\mathcal{I}$ locations (3,143 U.S. counties), two types of agents:
 - **Workers:** mobile, choose location, earn wages, consume goods and housing
 - **Capitalists:** immobile, own local capital, make investment decisions
- Global temperature $T_t - T^P = \epsilon(z_t + T_t^D)$ where T_t^D is a deterministic climate path and z_t captures natural variability; ϵ is a scaling parameter
- Location-specific **damage functions** depend on temperature:

$$\chi_{it} = \chi_i(T_t - T^P), \quad a_{it} = a_i(T_t - T^P), \quad \delta_{it} = \delta_i(T_t - T^P)$$

affecting productivity, amenities, and capital depreciation respectively

Workers (Bilal & Rossi-Hansberg 2023, §2.1, §2.5)

- Workers have CRRA preferences over amenities A_{it} and consumption C :

$$A_{it} + u(C) = A_{it} + \frac{C^{1-\gamma} - 1}{1-\gamma}, \quad C = \left(\frac{c}{1-\beta} \right)^{1-\beta} \left(\frac{h}{\beta} \right)^{\beta}$$

- At Poisson rate μ , workers draw T1EV preference shocks (scale ν) and may migrate, paying bilateral cost τ_{ij}
- The **worker HJB** for the value of residing in location i (eq. 3):

$$\rho V_{it} = U_i(a_{it}, \chi_{it}, K_{it}, N_{it}) + \mathcal{M}_i[V] + \frac{\mathbb{E}[d_t V_{it}]}{dt}$$

where $\mathcal{M}_i[V] \equiv \mu \left(\frac{1}{\nu} \log \left(\sum_j e^{\nu(V_{jt} - \tau_{ij})} \right) - V_{it} \right)$ is the migration option value

Migration Shares and the KFE (eq. 4–5)

- **Migration shares** from T1EV (eq. 4):

$$m_{ij}(V) = \frac{e^{\nu(V_j - \tau_{ij})}}{\sum_k e^{\nu(V_k - \tau_{ik})}}$$

- The **Kolmogorov Forward Equation** for population (eq. 5):

$$\frac{dN_{it}}{dt} = \mu \left(\sum_k m_{ki}(V_t) N_{kt} - N_{it} \right) \equiv \mu (m^*(V_t) - \text{Id}) N_t |_i$$

where m^* denotes the transpose of the migration share matrix m

- Define $M^*(V) = \mu(m^*(V) - \text{Id})$ and $M(V) = \mu(m(V) - \text{Id})$, so:

$$\frac{dN_t}{dt} = M^*(V) N_t$$

M acts on values (backward), M^* acts on populations (forward) — the adjoint pair

Capitalists and Capital (eq. 6–8)

- Capitalists are immobile, risk-neutral, own local capital K_{it}
- Q_{it} denotes the marginal value of capital in location i ; the **capitalist HJB** (eq. 7):

$$\rho Q_{it} = R_i(\chi_{it}, K_{it}, N_{it}) + \frac{c_{i0} Q_{it}^{1+\zeta}}{1+\zeta} - \delta_{it} Q_{it} + \frac{\mathbb{E}_t[d_t Q_{it}]}{dt}$$

where R_i is the return on capital and ζ governs investment adjustment costs

- Optimal investment: $I_{it}^* = c_{i0} Q_{it}^\zeta K_{it}$
- The **law of motion for capital** (eq. 8):

$$\frac{dK_{it}}{dt} = (c_{i0} Q_{it}^\zeta - \Delta_{it}) K_{it}$$

where $\Delta_{it} = \Delta_i + \delta_{it}$ is the total depreciation rate (long-run + climate-driven)

Steady State (§2.7, eq. 9–12)

At baseline temperatures ($\epsilon = 0$), all time derivatives vanish:

- **Worker HJB** (eq. 9): $\rho V_i^{SS} = U_i(0, 0, K_i^{SS}, N_i^{SS}) + \mathcal{M}_i[V^{SS}]$
- **Capitalist HJB** (eq. 10): $\rho Q_i^{SS} = R_i(0, K_i^{SS}, N_i^{SS}) - \frac{c_{i0}(Q_i^{SS})^{1+\zeta}}{1+\zeta}$
- **No net migration** (eq. 11): $0 = M^*(V^{SS}) N^{SS}$
- **Stable capital** (eq. 12): $0 = c_{i0}(Q_i^{SS})^\zeta - \Delta_i$

The aggregate state variables are $N_t = \{N_{it}\}_i$ and $K_t = \{K_{it}\}_i$: a total of $2\mathcal{I}$ state variables. With $\mathcal{I} = 3,143$ counties, this yields over 6,000 coupled state variables — the source of the computational challenge.

Strategy (§3.1)

THE CHALLENGE:

- The distributions N_t and K_t determine local prices (wages, rents, returns)
- These distributions evolve according to the KFE (eq. 5) and capital law of motion (eq. 8)
- The laws of motion depend on value functions V and Q from the HJB equations (eq. 3, 7)
- This creates a coupled forward-backward system in $> 6,000$ dimensions

THE MASTER EQUATION APPROACH (TWO STEPS):

- 1 Merge the individual decision problems with the laws of motion into a single Bellman equation: the **Master Equation**
- 2 Take a first-order perturbation of the Master Equation in ϵ around the steady state: the **FAME**

Step 1: Change of Variables (§3.2, eq. 13)

- Same idea as the simple case: rewrite $V_{it} = V_{it}(z_t, N_t, K_t)$, indexing by distributions rather than calendar time. The value depends on time only through the state variables (z, N, K) and the deterministic temperature trend T_t^D .

- Apply the chain rule to decompose the continuation value:

$$\frac{\mathbb{E}_{it}[d_t V_{it}]}{dt} = \underbrace{\frac{\partial V_{it}}{\partial t}}_{\text{change in } T_t^D} + \underbrace{\mathcal{A}_\epsilon(z)[V_{it}]}_{\text{change in } z_t \text{ (continuation value)}} + \underbrace{\sum_j \frac{\partial V_{it}}{\partial N_j} \frac{dN_{jt}}{dt}}_{\text{change in } N_t \text{ (chain rule)}} + \underbrace{\sum_j \frac{\partial V_{it}}{\partial K_j} \frac{dK_{jt}}{dt}}_{\text{change in } K_t \text{ (chain rule)}}$$

- $\mathcal{A}_\epsilon(z)$: the generator of z_t (captures natural climate variability)
- The last two terms are the chain rule terms — exactly as in the simple case, but now applied to both N and K . They encode how changes in the population and capital distributions affect the value of workers in location i

Step 1: The Master Equation for Workers (eq. 14)

Now substitute the laws of motion into the chain rule (Step 2 in the simple case), and plug the result into the worker HJB (Step 3). The KFE gives $\dot{N} = M^*(V)N$; the capital law of motion gives

$$\dot{K}_{it} = (c_{i0}Q_{it}^{\zeta} - \Delta_{it})K_{it}:$$

$$\begin{aligned} \rho V_{it}(z, N, K) = & \underbrace{U_i(\cdot)}_{\text{flow payoff}} + \underbrace{\mathcal{M}_i[V]}_{\text{migration option}} \\ & + \underbrace{\sum_j \frac{\partial V_{it}}{\partial N_j} (M^*(V)N_t)_j + \sum_j \frac{\partial V_{it}}{\partial K_j} (c_{j0}Q_{jt}^{\zeta} - \Delta_{jt})K_j}_{\text{continuation values from aggregate changes}} + \mathcal{A}_{\epsilon}(z)[V_{it}] + \frac{\partial V_i}{\partial t} \end{aligned}$$

- Same structure as the simple case, but with two chain rule terms (for N and K) plus climate terms ($\mathcal{A}_{\epsilon}$, $\partial V/\partial t$). The KFE and capital law of motion are absorbed.
- The worker ME depends on Q because capitalist investment affects prices and worker welfare.

Step 1: The Master Equation for Capitalists (eq. 15)

The same logic applies to the capitalist value function $Q_{it}(z, N, K)$:

$$\begin{aligned} \rho Q_{it}(z, N, K) = & \underbrace{R_i(\cdot) + \frac{c_{i0} Q_i^{1+\zeta}}{1+\zeta} - \Delta_{it} Q_{it}}_{\text{flow payoff + own investment}} \\ & + \underbrace{\sum_j \frac{\partial Q_{it}}{\partial N_j} (M^*(V) N_t)_j + \sum_j \frac{\partial Q_{it}}{\partial K_j} (c_{j0} Q_{jt}^\zeta - \Delta_{jt}) K_j}_{\text{continuation values from aggregate changes}} + \mathcal{A}_\epsilon(z)[Q_{it}] + \frac{\partial Q_i}{\partial t} \end{aligned}$$

- Together, equations (14) and (15) form a system of two coupled Master Equations
- **Proposition 1** (Bilal & Rossi-Hansberg): A solution (V, Q) to the Master Equations, together with $\{N_t, K_t\}_t$ satisfying the KFE and capital law of motion, constitutes an equilibrium

Step 2: First-Order Taylor Expansion (§3.3, eq. 16)

Just as in the simple case, linearize around steady state. Now there are more coefficients because there are two state vectors (N and K) and climate terms. Set $N = N^{SS} + \epsilon n$ and $K = K^{SS} + \epsilon k$:

$$V_i(z, N^{SS} + \epsilon n, K^{SS} + \epsilon k) = V_i^{SS} + \epsilon \left(\sum_j v_{ij}^N n_j + \sum_j v_{ij}^K k_j + v_i^Z z + v_{it}^T \right) + O(\epsilon^2)$$

$$Q_i(z, N^{SS} + \epsilon n, K^{SS} + \epsilon k) = Q_i^{SS} + \epsilon \left(\sum_j q_{ij}^N n_j + \sum_j q_{ij}^K k_j + q_i^Z z + q_{it}^T \right) + O(\epsilon^2)$$

- v_{ij}^N, v_{ij}^K : how worker values respond to population and capital changes (**deterministic impulse values**)
- v_i^Z : response to natural climate variability (**stochastic impulse value**)
- v_{it}^T : response to the deterministic temperature trend (**trend impulse value**)
- Analogous objects q^N, q^K, q^Z, q^T for capitalists

From One Expansion to Three FAMEs

Substitute the expansion into the Master Equations and collect first-order terms on each perturbation source separately:

COEFFICIENTS ON $n_j, k_j \longrightarrow$ **Deterministic FAME** (Prop. 2):

$$\rho v^d = D^d + M v^d + v^d M^* + v^d P^d v^d$$

COEFFICIENTS ON $T_t^D \longrightarrow$ **Trend FAME** (Prop. 3):

$$\rho v_t^T = D_t^T + M v_t^T + \frac{\partial v_t^T}{\partial t} + v^d P^d v_t^T$$

COEFFICIENTS ON $z_t \longrightarrow$ **Stochastic FAME** (Prop. 3):

$$\rho v^Z = D^Z + M v^Z + v^d P^d v^Z$$

Block-recursivity: v^d appears in the other two but not vice versa. Solve v^d first (nonlinear); then v_t^T and v^Z are each **linear** given v^d .

Step 2: The Deterministic FAME (Proposition 2)

Substitute the Taylor expansion into the Master Equations and collect terms on n_j and k_j . Stack the impulse values into $2\mathcal{I} \times 2\mathcal{I}$ block matrices:

$$v^d \equiv \begin{pmatrix} v^N & v^K \\ q^N & q^K \end{pmatrix}$$

- Compare with the simple-case FAME ($\rho v = D + Mv + vM^* + vGv$). The structure is identical, but now v^d is a $2\mathcal{I} \times 2\mathcal{I}$ block matrix capturing responses to both N and K :

$$\rho v^d = D^d + M v^d + v^d M^* + v^d P^d v^d$$

where:

$$D^d = \begin{pmatrix} -D_{UN} & D_{UK} \\ -D_{RN} & D_{RK} \end{pmatrix}, \quad M = \begin{pmatrix} M & 0 \\ 0 & 0 \end{pmatrix}, \quad P^d = \begin{pmatrix} G & 0 \\ 0 & D_{IQ} \end{pmatrix}$$

Interpreting the Deterministic FAME

$$\rho v^d = \underbrace{D^d}_{(1)} + \underbrace{M v^d}_{(2)} + \underbrace{v^d M^*}_{(3)} + \underbrace{v^d P^d v^d}_{(4)}$$

- ➊ D^d : **direct price impact**. When N or K change, prices shift. D_{UN} captures the congestion effect on wages; D_{RK} captures the effect on capital returns.
- ➋ $M v^d$: **partial equilibrium migration**. Workers evaluate the migration option using the steady-state transition matrix M . This holds by the envelope condition.
- ➌ $v^d M^*$: **first general equilibrium force**. An additional worker in j behaves like any other worker and migrates according to M^* . The adjoint M^* right-multiplies v^d .
- ➍ $v^d P^d v^d$: **second general equilibrium force**. An additional worker or capital changes prices, which alters migration and investment decisions, feeding back through LOM.

Step 2: The Trend FAME (Proposition 3)

Collect coefficients on the deterministic temperature path T_t^D . Stack $v_t^T \equiv (v_t^T, q_t^T)^T$:

$$\rho v_t^T = D_t^T + M v_t^T + \frac{\partial v_t^T}{\partial t} + v^d P^d v_t^T$$

- $D_t^T = T_t^D \begin{pmatrix} D_{UT} - v^K D_{\Delta T} \\ -D_{QT} - q^K D_{\Delta T} \end{pmatrix}$: the **direct climate damage**, combining effects on amenities (a_{i1}), productivity (χ_{i1}), and capital depreciation (δ_{i1})
- $M v_t^T$: migration response to climate damages (via steady-state M)
- $\frac{\partial v_t^T}{\partial t}$: the trend FAME is an **ODE in calendar time** — climate change is non-stationary
- $v^d P^d v_t^T$: general equilibrium feedback through the law of motion

BLOCK-RECURSIVITY: v^d is solved first; then v_t^T (and v^Z) can be computed given v^d .

Step 3: Transitional Dynamics (Proposition 4)

Given impulse values v^d , v^T , v^Z , the transitional dynamics of (n_t, k_t) satisfy:

$$\frac{d}{dt} \begin{pmatrix} n_t \\ k_t \end{pmatrix} = (M^* + P^d) \begin{pmatrix} n_t \\ k_t \end{pmatrix} + P^d(v_t^T + z_t v^Z) - (T_t^D + z_t) \begin{pmatrix} 0 \\ D_{\Delta T} \end{pmatrix}$$

- This is a **linear ODE**: given any path of shocks $\{z_t, T_t^D\}_t$, the population and capital distributions can be integrated forward directly
- The FAME inherits block-recursivity from the Master Equation: it is the only fixed point that must be computed
- No additional price or distributional fixed point remains — all fixed points are absorbed into the Master Equation and hence the FAME

Summary of the FAME Procedure

PROCEDURE (BILAL & ROSSI-HANSBERG 2023, §3):

- ➊ **Steady state:** Solve the nonlinear system (eq. 9–12) for V^{SS} , Q^{SS} , N^{SS} , K^{SS}
- ➋ **Master Equation:** Combine HJB equations with KFE and capital law of motion into recursive equations (14)–(15) in (V, Q, N, K)
- ➌ **Deterministic FAME:** Solve the Sylvester equation for v^d (Proposition 2)
- ➍ **Trend/Stochastic FAME:** Given v^d , solve for v_t^T and v^Z (Proposition 3)
- ➎ **Transitional dynamics:** Integrate the linear ODE for (n_t, k_t) forward (Proposition 4)

RESULT: A tractable characterization of transitional dynamics across 3,143 counties with > 6,000 state variables — without iterating on coupled forward-backward time paths.

Why the Method Matters for This Application

- WITHOUT FAME:

- Solving a model with 6,000+ coupled state variables via forward-backward iteration would require prohibitive computational resources
- Standard methods (e.g., Caliendo, Dvorkin, Parro 2019) are limited to approximately 50 locations

- WITH FAME:

- Perturbation reduces the problem to solving matrix equations
- The method handles 3,143 counties on standard hardware
- The near-linearity result: first-order perturbation is highly accurate for conventional-magnitude climate shocks

References

- Bilal (2023), “Solving Heterogeneous Agent Models with the Master Equation”
- Bilal & Rossi-Hansberg (2023), “Anticipating Climate Change Across the United States”
- Caliendo, Dvorkin, Parro (2019), “Trade and Labor Market Dynamics: General Equilibrium Analysis of the China Trade Shock”
- Desmet, Nagy, Rossi-Hansberg (2018), “The Geography of Development”
- Kleinman, Liu, Redding (2023), “Dynamic Spatial General Equilibrium”
- Stokey (2008), “The Economics of Inaction”